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Enabling boundary spanners in public–private collaboration: The impact of support and role autonomy on reducing role stress

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Abstract

Boundary spanners are pivotal in developing effective public–private collaboration in public infrastructure projects. As boundary spanners have to account for different interests, perspectives, and ways of working in public–private collaborations, engaging in boundary-spanning activities often comes with role stress, which can negatively impact their job performance. However, despite the significant levels of role stress associated with boundary spanning, there is a dearth of empirical studies on the role stress of boundary spanning public managers and how it can be reduced. In this study, we empirically investigate the reciprocal relationship between role stressors and boundary spanning and test if supporting organizational conditions can alleviate role stressors. The findings show that especially role conflict is detrimental for boundary-spanning activities. We further find that top-management support can diminish both role conflict and role ambiguity and that co-worker support can help in reducing the role ambiguity of boundary spanning public managers.

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1 | INTRODUCTION

Public-private collaboration has the potential to improve the quality and operational efficiency of public projects that are implemented in complex environments, such as in the case of the delivery of public infrastructure projects (Kwak et al., 2009). This can be achieved by joining efforts to solve issues that a single organization could find difficulty in resolving by itself (Ansell & Gash, 2007). Public management literature suggests, however, that barriers to developing good collaboration between public and private partners can arise from conflicting perceptions between the actors involved (Klijn & Teisman, 2003). Boundary spanners play a key role in implementing good collaboration by developing, maintaining, and coordinating the collaboration between public and private sector partners in public infrastructure projects (Noble & Jones, 2006; Van Meerkerk & Edelenbos, 2018a; Williams, 2012). They help to capture the benefits of collaboration, for instance by developing trust (Delozier & Burbach, 2021). The level of trust can be seen as a determinant of the quality of the collaboration (Warsen et al., 2019); and boundary spanners could aid in developing trust by searching for common ground, building sustainable relationships, and by communicating across boundaries in order to understand the partner's needs through active listening (Van Meerkerk & Edelenbos, 2018a; Williams, 2012). Some of the other activities they undertake include information processing (Marrone, 2010), external representation of the organization (Ancona & Caldwell, 1992), recognizing opportunities, handling setbacks by managing conflicts (Delozier & Burbach, 2021; Williams, 2012), and transferring and managing information flows (Tushman & Scanlan, 1981; Williams, 2012).

Although some boundary-spanning activities may be explicitly part of a public manager's job description or organizational role, the demand, and extent of engaging in these activities is often not specified in a job profile but rather is based on one's own perception of the project requirements (Van Meerkerk & Edelenbos, 2018a). This is challenging because boundary spanning public managers have to adapt to continuously changing project environments, engaging in collective sensemaking with different stakeholder groups in public-private collaborations (Noble & Jones, 2006). Their engagement in boundary-spanning activities and dealing with various intra-organizational and extra-organizational demands and interests—which is particularly pertinent in the case of public-private collaboration, where public and private values easily conflict with each other (e.g., Klijn & Teisman, 2003; Weihe, 2008)—often causes boundary spanners to experience high levels of role stress (as compared to other organizational employees), as highlighted in business administration research (e.g., Nygaard & Dahlstrom, 2002; Singh, 1998; Woisetschlager et al., 2016). Role stress results from a mismatch between an individual's expectation and operation of a role. Two key and well-known role stressors are role conflict, which may occur due to conflicting role demands (Stamper & Johlke, 2003), and role ambiguity, which is the lack of clarity regarding the expectations from one's role (Rizzo et al., 1970). These role stressors are experienced by persons internally, which can negatively impact the public manager's job performance and the job satisfaction of boundary spanners, their boundary-spanning activities, as well as their organizational commitment (Bettencourt & Brown, 2003; Chebat & Kollias, 2000; Crosno et al., 2009; Stamper & Johlke, 2003). There is substantial evidence that, in general, public managers are confronted with conflicting or confusing demands, triggering role stressors in public management (Pandey & Wright, 2006). This is especially the case for public managers who operate in public-private partnerships or networks and in their home organization simultaneously (Davis & Stazyk, 2016; Weihe, 2008). The relationships between boundary spanning and role stress have generally been studied in two ways. Some studies focus on the negative effects of role stressors on a boundary spanner's job performance (including their spanning activities), job satisfaction, and/or organizational commitment (Chebat & Kollias, 2000; Crosno et al., 2009; Stamper & Johlke, 2003). Other studies focus on the effects of engaging in boundary-spanning activities on the experience of role stress (e.g., Nygaard & Dahlstrom, 2002). Most studies have either examined the effect of role stressors on boundary spanning or investigated the opposite effect of boundary spanning on role stressors (Davis & Stazyk, 2016; Robertson, 1995). However, very few studies have simultaneously investigated these relationships, to see if there is in fact a reciprocal relationship between role stressors and boundary spanning.

Although there is increasing research on the importance of relational management and relational governance mechanisms for enhancing the performance of public–private collaborations in public infrastructure projects (e.g., Warsen et al., 2019), attention to the challenging role of boundary spanners in managing these collaborations, and specifically how role stressors could be coped with, is scarce (cf. Noble & Jones, 2006; Van Meerkerk & Edelenbos, 2018a). This is an under-researched topic in the public administration literature (Van Meerkerk & Edelenbos, 2018a; cf. Davis & Stazyk, 2016).

The aim of this study is twofold. First, to test the mutual relationship between two key role stressors—that is, role conflict and role ambiguity—on one hand, and engagement in boundary-spanning activities on the other. Second, to test the impact of the three supporting organizational conditions on role conflict and role ambiguity. Drawing on social psychology, business administration, and public administration, scholars suggested that top-management support, co-worker support, and role autonomy are three important organizational conditions that can facilitate boundary-spanning activities by reducing boundary spanners' role stress (Chebat & Kollias, 2000; Eldor, 2018; Perrone et al., 2003; Sloan, 2017). Translating them to the context of public–private collaboration, we aim to put these factors to the test. A suitable empirical setting to study these relationships concerns the infrastructure development sector, as public–private collaboration is a popular policy instrument for developing public infrastructure projects (Busscher et al., 2022; Warsen et al., 2019). The research data are composed of 158 survey responses from public managers involved in developing public infrastructure projects in the Netherlands.

2 | THEORETICAL FRAMEWORK AND HYPOTHESES

We first discuss boundary-spanning activities and their contribution to public–private collaborations. We then discuss role theory and two key role stressors often experienced by boundary spanners—that is, role ambiguity and role conflict—and develop hypotheses about how they relate to boundary-spanning activities. We subsequently discuss different supporting organizational conditions—top-management support, co-worker support, and role autonomy—and how they could reduce role stressors.

2.1 | Boundary-spanning activities in public–private collaboration in infrastructure projects

Boundary spanners are essential in establishing and maintaining collaborative relationships (Ancona & Caldwell, 1992). They can be defined as “people who proactively scan the organizational environment, employ activities to cross organizational or institutional boundaries, generate and mediate the information flow and coordinate between their ‘home’ organization or organizational unit and its environment, and connect processes and actors across these boundaries” (Van Meerkerk & Edelenbos, 2018a, p. 58). Based on role typologies developed in the literature, different types of boundary-spanning activities can be distinguished (Van Meerkerk & Edelenbos, 2018a; Williams, 2012). The first one is information exchange and knowledge sharing. Boundary spanners manage information exchange by collecting, processing, and translating relevant information across boundaries (Tushman & Scanlan, 1981). They mediate the flow of information and prioritize information, which can foster informed decision-making processes (Delozier & Burbach, 2021). The second is relational activities. Relational activities involve developing the personal network, which also facilitates knowledge and information exchange across boundaries (Ancona & Caldwell, 1992; Brion et al., 2012). Third is the mediation and facilitation of collaboration. Boundary spanning requires an ability to interact with external parties, that may have diverging interests to that of the home organization, and to redirect the opinions of the partners to the benefit of the project (Lundberg, 2013). A fourth type of activity concerns coordination and negotiation between the home organization and partner organizations. This allows for tighter coupling and smooth running of cross-boundary collaboration (Ancona & Caldwell, 1992;

Williams, 2012). Undertaking this activity helps to organize the logistics of the collaboration, by coordinating tasks and meetings, taking charge of administration, and monitoring progress (Williams, 2012).

Building on the framework of Van Meerkerk and Edelenbos (2018a) from the public management and governance literature, we examine the role of organizational conditions which could facilitate boundary spanners in dealing with role stressors that affect their boundary-spanning activities. This is visualized in the conceptual model (see Figure 1). In the next section, we go deeper into the different parts of the conceptual model.

2.2 | Role theory and role stressors

Role theory conceptualizes organizations as interactive social systems, where the actions of employees are constrained and directed by the expectations of the others in the collaborative environment (Kahn et al., 1964; Katz & Kahn, 1978). Therefore, the behavior of boundary spanners is determined by influences, expectations, and pressures from groups and members of their own organization, but also from groups interacting with the organization, such as members from collaborating organizations (Van Meerkerk & Edelenbos, 2018a). Because these different groups have different interests and values, boundary spanners are likely to experience uncertainty and/or conflicting expectations about how they should perform their role (Klijn & Teisman, 2003). For instance, public managers in public–private collaborations often have a “best for project” goal that prioritizes collaboration and project outcomes, but at the same time, they have a position as clients and auditors of the private partner organizations (see e.g., Verweij, 2012).

These uncertainties or conflicts can lead to the development of role stressors, which commonly occur when there is an incongruence between the expectation and the execution of a role. Role conflict and role ambiguity are seen as the most critical stressors for boundary spanners (Bettencourt & Brown, 2003; Crosno et al., 2009; Wang et al., 2020; Wu et al., 2019). According to role theory, an increase in role conflict or role ambiguity could lead to a decrease in (job) performance (Kahn et al., 1964; Rizzo et al., 1970). In addition, role stressors are found to create tension and dissatisfaction for boundary spanning employees and are found to be the cause of poor judgment, bad decisions, mismanagement of resources, and overall low morale (Lysons et al., 1988), in turn causing them to experience role stressors (Robertson, 1995).

Role conflict refers to incompatible expectations and demands associated with one's work role (Rizzo et al., 1970), for example, incompatible expectations coming from supervisors, co-workers, and partners in the collaboration (Wu et al., 2019). Role ambiguity occurs when an employee lacks the information needed to effectively enact

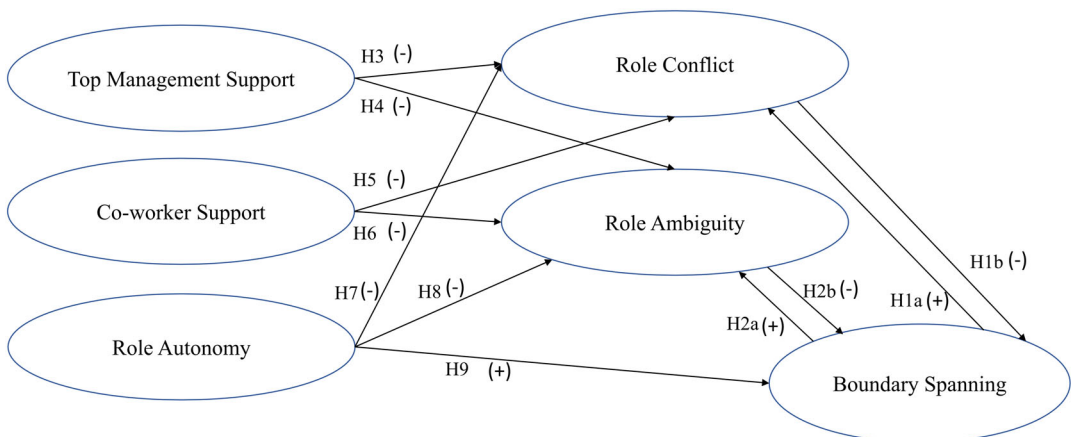


FIGURE 1 Conceptual model. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/padm.12927)]

their role (Kahn et al., 1964), or when there is uncertainty about expectations from constituencies (Rizzo et al., 1970). Boundary spanners often spend much of their time interacting with stakeholders, addressing their variable, complex, and distinctive needs and interests (Stamper & Johlke, 2003). Although these stakeholders can be manifold—for example, citizens, customers, and/or clients—in this study, we focus only on the public and private organizations as stakeholders, as they are directly involved in the collaboration, between whom also often multiple and contrasting role expectations exist (Van Meerkerk & Edelenbos, 2018a). As a consequence, boundary spanners are likely to experience relatively high levels of role conflict and role ambiguity. This leads to the following hypotheses:

Hypothesis 1a. (H1a): Public managers engaging in more boundary-spanning activities in public–private collaboration in public infrastructure projects experience more role conflict.

Hypothesis 2a. (H2a): Public managers engaging in more boundary-spanning activities in public–private collaboration in public infrastructure projects experience more role ambiguity.

Previous research has established that boundary spanners are vulnerable to role conflict (Kahn et al., 1964), and that role conflict (through lower job satisfaction) negatively affects boundary spanning behavior (Bettencourt & Brown, 2003). Role conflict arising from conflicting demands could even result in burnout (Crosno et al., 2009). This means that role conflict becomes a hindrance in accomplishing boundary-spanning activities and affects the boundary spanner's performance. In the long run, prolonged exposure to role conflict might lead to withdrawal from one's job. This is undesirable, as changes in managerial positions in public–private collaborations may derogate relationships built and may result in lower project performance (Busscher et al., 2022). In line with this, we hypothesize that role conflict negatively impacts boundary spanning because the ability to perform boundary-spanning activities decreases when dealing with stress due to role conflict (Cohen & Wills, 1985):

Hypothesis 1b. (H1b): Role conflict decreases the extent of boundary-spanning activities undertaken by public managers in public–private collaboration in public infrastructure projects.

Existing studies on role ambiguity indicate that its influence on boundary spanning is mixed (i.e., positive, negative, or neutral influence can all be found) but it appears it is predominantly a hindering or a negative stressor (Babin & Boles, 1996; Singh, 1998; Woisetschlager et al., 2016). Often, role ambiguity is caused either by incongruent role expectations (Rizzo et al., 1970), or by discrepancy in available information (Kahn et al., 1964). The consequence of role ambiguity is associated with negative emotions, as it could lead to experiencing lower levels of self-confidence, increase in anxiety, and depression (Lee & Ashforth, 1996). In a public–private collaboration, it is likely that public project managers will experience role ambiguity because of a limited understanding about their role, due to insufficient interaction or information when working across boundaries (Wu et al., 2019). Role ambiguity could be seen as a threat that lowers job satisfaction (Crosno et al., 2009; Kahn et al., 1964), which can reduce the extent of boundary-spanning activities being performed in a public–private collaboration (Nygaard & Dahlstrom, 2002). This leads to the following hypothesis:

Hypothesis 2b. (H2b): Role ambiguity decreases the extent of boundary-spanning activities undertaken by public managers in public–private collaboration in public infrastructure projects.

2.3 | Top-management support

Top-management support is the employee's perception of the care from the employer—by catering to the employee's needs, such as organizational membership, esteem, and role status—and the reward for their boundary-

spanning activities (Rhoades & Eisenberger, 2002). Top-management support is found to have a significant negative association with role stressors (Kurtessis et al., 2017): support in terms of encouragement and compassion was found to decrease stress among public managers (Eldor, 2018). The employee's perception of top-management support is developed through multiple interactions with their employer over time (Rhoades & Eisenberger, 2002), and by providing crucial resources for facilitating the execution of tasks (Babin & Boles, 1996). Supporting the boundary spanners is beneficial for an organization; support from top-management has, for instance, shown to enhance project performance (Eskered et al., 2017). By providing a conducive working environment, the self-efficacy of the boundary spanner is enhanced, thereby also increasing the ability to handle role stressors (Kurtessis et al., 2017). In particular, role ambiguity can be alleviated by the support provided in terms of eliminating unnecessary workload and being clearer about the job expectations (Hobfoll, 1989). Top-management support can be seen as a counteracting factor against role stressors and a steady support from top-management allows the boundary spanners to cope with role conflict and ambiguity when engaging in boundary-spanning activities. This leads to the following hypotheses:

Hypothesis 3. (H3): Top-management support reduces role conflict of public managers engaged in public-private collaboration in public infrastructure projects.

Hypothesis 4. (H4): Top-management support reduces role ambiguity of public managers engaged in public-private collaboration in public infrastructure projects.

2.4 | Co-worker support

Research shows that support from co-workers can help in dealing with work-related issues (Nguyen & Tuan, 2022), and diminish the negative effects of role stressors through emotional support (Cohen & Wills, 1985). Downplaying the conflicting demands that boundary spanners face may be experienced as a lack of acknowledgment for their efforts, which would lessen the motivation to deal with role conflict and role ambiguity (Babin & Boles, 1996; Sloan, 2017). Instead, supporting them emotionally and assisting them in completing tasks can empower the boundary spanner (Sloan, 2017). The interactions in their working environment can positively influence their behavior (Hammer et al., 2004). Having co-worker support can reduce stress (Babin & Boles, 1996), by providing information on how they handled similar situations in the past, and this develops trust and positive attitudes toward work engagements (Nguyen & Tuan, 2022). On this basis, the following hypotheses can be developed:

Hypothesis 5. (H5): Co-worker support reduces role conflict of public managers engaged in public-private collaboration in public infrastructure projects.

Hypothesis 6. (H6): Co-worker support reduces role ambiguity of public managers engaged in public-private collaboration in public infrastructure projects.

2.5 | Role autonomy

Earlier studies showed that having autonomy can help in coping with role stress and can increase job performance (Singh, 1993; Singh, 1998). Role autonomy is experiencing the freedom and flexibility to interpret the role and have the discretionary power to proceed as an individual sees fit under specific circumstances (Nygaard & Dahlstrom, 2002; Perrone et al., 2003). For a boundary spanner, the uncertainty associated with operating across boundaries can induce role stressors (Lyonski et al., 1988). When the desired policy outcome is complex, micro-controlling an employee's actions could potentially hinder the task performance (Pandey & Wright, 2006). With

regard to boundary spanning, loss of control on how a task is performed can be linked to creating an ambiguity in role expectation. Having the freedom to do the tasks in one's own way would reduce the pressure to conform to procedures, which might contribute to reducing role conflict and role ambiguity. This leads to the following hypotheses:

Hypothesis 7. (H7): Role autonomy reduces role conflict of public managers engaged in public–private collaboration in public infrastructure projects.

Hypothesis 8. (H8): Role autonomy reduces role ambiguity of public managers engaged in public–private collaboration in public infrastructure projects.

Organizations can benefit from recognizing the role of boundary spanners, providing them the room to maneuver to deal with problems, and valuing the outcomes generated by the external orientation of boundary spanners (Nygaard & Dahlstrom, 2002). In a study involving stakeholders from integrated water resource management, the centrality of a boundary spanner in a multi-stakeholder network was acknowledged by the participants (Delozier & Burbach, 2021; Van den Brink et al., 2019). An essential trait for that individual was to experience a certain degree of autonomy from their own organization and this helps boundary spanners to interact in an unbiased manner between stakeholders (Delozier & Burbach, 2021). This helps the boundary spanner with a stronger position to better manage the relationships with collaborators (Perrone et al., 2003). In other words, having a degree of autonomy might not only reduce role stressors experienced by the boundary spanner, but also have a direct effect on boundary spanning itself (Brion et al., 2012). Therefore, we hypothesize that:

Hypothesis 9. (H9): Role autonomy increases boundary-spanning activities of public managers engaged in public–private collaboration in public infrastructure projects.

3 | DATA AND METHODS

3.1 | Sample and data collection

Data were collected from lead public managers through a survey questionnaire (Satheesh et al., 2022). The public managers were selected based on their involvement in the implementation phase of infrastructure projects. These managers are affiliated with Rijkswaterstaat, the executive agency of the Ministry of Infrastructure and Water Management in the Netherlands. The questionnaire included items derived from previous scales and new items developed based on the literature. The survey was developed in English and then translated to Dutch, as the respondent pool comprised of Dutch speakers. Subsequently, the questionnaire was translated back to English for the analysis. A pilot survey was distributed among Rijkswaterstaat managers who were not later invited as participants in the survey, and the received suggestions were used to refine the measurement items. The survey was online and data were anonymized.

The surveyed projects were selected based on two criteria: they were major infrastructure development projects (major highways, water works, bridges, and/or tunnels) and the contract type for partnering with the private sector was either Design-and-Construct (D&C) or a Design-Build-Finance-Maintain (DBFM) (see more about the contract types in Verweij & Van Meerkerk, 2020). Important distinguishing elements of DBFM—compared to D&C—are that DBFM includes private financing, performance-dependent payments based on services, the inclusion of infrastructure maintenance, and increased risk transfer to the private consortium (Koppenjan et al., 2022). Project records were available for projects that started between 2006 and 2018. Minor maintenance work projects, knowledge projects or programs, and so on were excluded because these involve different stakeholder networks, procurement

processes, and management dynamics (Verweij & Van Meerkerk, 2020). This created a project pool of 68 projects and 406 potential respondents with 308 unique respondents. Respondents were asked to fill in the survey for a specific project. If they were involved in more than one project, they were asked to fill out the survey for a maximum of two projects (the most recent ones in which they were involved). After eliminating 14 invalid email addresses, the sample size was 392. The survey was conducted between October 2020 and January 2021. A total of 171 public managers from Rijkswaterstaat responded (the response rate after two follow-ups was 43.6%), out of which 13 responses were deleted due to a completion rate of less than 35%. This resulted in a dataset with 158 responses (Satheesh et al., 2022). To determine the project and respondent characteristics, respondents were asked to fill in personal demographics, their role in the project team, the contract type, to list the amount of stakeholders involved (i.e., stakeholder complexity), and to list the related spatial developments—road construction, improving waterway network, ensuring water quality and safety, housing, development of industrial, commercial or recreational area, and landscaping—that were part of the project (i.e., spatial complexity) (cf. Verweij et al., 2013). The descriptive statistics of the sample is shown in Table 1.

To test for bias in nonresponse, we examined the differences between the first and the last 79 responses. The test showed that there were no significant differences between the two groups in demographic characteristics such as age, years of experience, and gender, nor in project characteristics such as contract type, infrastructure system, stakeholder complexity, and spatial complexity ($p > 0.05$).

There is the possibility of having common method bias in the tested relationships, as the test is self-reported. To limit this bias, we had taken research design remedies (George & Pandey, 2017), such as contextualizing the study items to fit the framework of public managers in public–private collaboration in public infrastructure projects and ensuring anonymity of the survey respondents to prevent social desirability bias (Podsakoff et al., 2003). Harman's single factor test was further used to test for variations in responses that can be attributed to the measurement method leading to systemic measurement errors. The results showed that a single factor could account for only 29% of the variance, which is much less than the threshold of 50% (Harman, 1976). This is an indication that it is unlikely that common method bias might be a threat to this study. In addition, it can be noted from the correlation matrix that both positive and negative correlations exist between the factors, as well as nonsignificant correlations. This demonstrates that common method bias is probably not a major concern in this dataset (George & Pandey, 2017).

3.2 | Measurement of variables

All the items for this study, presented in Table 2, were measured using a 7-point Likert scale. The choices for the boundary spanning measurement scale ranged from 1 (“very little involvement”) to 7 (“involvement to a great extent”). The items for the organizational conditions investigated how they experienced their role within the collaboration of the specified project and it was measured by checking the extent to which the respondent was in agreement with the item scaling from 1 (“strongly disagree”) to 7 (“strongly agree”).

Role conflict and role ambiguity were also measured with 1 indicating “strongly disagree” to 7 indicating “strongly agree.” Role ambiguity was conceptualized as a lack of clarity regarding the expectation of the role (Babin & Boles, 1996). The items for role ambiguity in Table 2 are reverse-coded, as the survey items measured the degree of clarity present in role definition.

3.2.1 | Control variables

Boundary spanning has been observed to be influenced by many characteristics, including years of work experience, personality traits, and personal competencies (Marrone et al., 2007; Tushman & Scanlan, 1981; Van Meerkerk & Edelenbos, 2018a). The demographic characteristics (age, years of experience, and gender) and project characteristics

TABLE 1 Summary of demographics and project characteristics distribution of the survey.

Variable	Categories	Frequency	Percentage
Demographics of the respondent			
Age group	24 years or younger	0	0
	25–34 years	2	1.26
	35–44 years	26	16.45
	45–54 years	77	48.73
	55–64 years	43	27.21
	65 years and older	10	6.33
Years of experience	1–5 years	11	6.96
	6–10 years	40	25.31
	11–20 years	70	44.30
	21–30 years	25	15.82
	More than 30 years	12	7.60
Gender	Male	130	82.27
	Female	25	15.82
	Other	3	1.91
Project characteristics			
Contract type	Design-and-Construct (D&C)	92	58.23
	Design-Build-Finance-Maintain (DBFM)	66	41.77
Infrastructure system	Road network	122	77.21
	Water network	36	22.78
Formal role	Project manager	24	15.19
	Contract manager	30	18.98
	Technical manager	28	17.72
	Project control manager	38	24.05
	Stakeholder manager	38	24.05
Spatial complexity (project scope)	1 spatial function	76	48.10
	2 spatial functions	42	26.58
	3 spatial functions	19	12.02
	4 spatial functions	12	7.59
	5 spatial functions	8	5.06
	6 spatial functions	1	0.63
Stakeholder complexity (number of stakeholders in the network)	1–10 stakeholders	51	32.27
	11–20 stakeholders	47	29.74
	21–30 stakeholders	22	13.92
	31–40 stakeholders	6	3.79
	41–50 stakeholders	9	5.69
	More than 50 stakeholders	23	14.55

TABLE 2 Measures and Validation.

Construct and items	Factor loading	Source
<i>Boundary Spanning Activity</i> (Cronbach's alpha (α) = 0.968; Composite reliability (CR) = 0.968; Average variance extracted (AVE) = 0.684) “During the project, to what extent have you been involved in...”		
Information collection and knowledge (IK) exchange activities		
• IK1: Exchanging of managerial information with the partner organization	0.829	Partly adapted from Yan et al. (2020), Ancona and Caldwell (1992), Henttonen et al. (2016), and Van Meerkerk and Edelenbos (2014)
• IK2: Exchanging technical knowledge and information with the partner organization	0.716	
• IK3: Exchanging and combining ideas with the partner organization	0.873	
Relational activities (RA)		
• RA1: Building and maintaining lasting relationships with the partner organization	0.903	Partly based on Walter et al. (2006) and Van Meerkerk and Edelenbos (2014)
• RA2: Organizing informal contact events with people from the partner organization	0.829	
• RA3: Developing personal relationships with people from the partner organization	0.758	
• RA4: Connecting people from the partner organization with people from my own organization	0.805	
Coordination and negotiation (CN) activities		
• CN1: Organizing formal meetings with the partner organization	0.809	Partly based on Walter et al. (2006), Ancona and Caldwell (1992), and Van Meerkerk and Edelenbos (2018b)
• CN2: Negotiating activities with the partner organization	0.810	
• CN3: Coordinating with contact persons of the contractor about how we can support each other for the benefit of the project	0.915	
• CN4: Coordinating tasks and roles between Rijkswaterstaat and the contractor	0.808	
Mediation and facilitation (MF) of collaboration		
• MF1: Mediating between the interests of Rijkswaterstaat and the interests of the contractor	0.824	Partly adapted from Ancona and Caldwell (1992) and Jehn and Mannix (2001)
• MF2: Facilitating collaboration between Rijkswaterstaat and the contractor	0.902	
• MF3: Stimulating consensus between Rijkswaterstaat and the contractor	0.857	
<i>Organizational factors</i>		
“In your role, to what extent did you experience the following conditions:...”		
Top-management support (TMP) (α = 0.854; Composite reliability (CR) = 0.857; AVE = 0.667)		
• TMP1: The management gave us good support when we needed it	0.755	Adapted from Van Meerkerk and Edelenbos (2018b)
• TMP2: The management made an explicit commitment to the project	0.894	
• TMP3: Management managed to generate attention and resources for the project	0.806	

TABLE 2 (Continued)

Construct and items	Factor loading	Source
Role autonomy (RAUTO) ($\alpha = 0.876$; CR = 0.875; AVE = 0.701)		
• RAUTO1: I had the freedom to fill in my role as I saw fit	0.864	Adapted from Tummers and Bekkers (2014)
• RAUTO2: In the project I had the space to determine which tasks had to be performed	0.810	
• RAUTO3: In the project I was able to make my own decisions	0.841	
Co-worker support (CWS) ($\alpha = 0.933$; CR = 0.935; AVE = 0.783)		
• CWS1: I was helped and supported by my colleagues	0.890	Adapted from Hammer et al. (2004)
• CWS2: I felt accepted in the team	0.922	
• CWS3: My colleagues understood when I had a bad day	0.802	
• CWS4: I felt comfortable with my colleagues	0.919	
Role stressors		
“In your role, to what extent did you experience the following conditions:...”		
Role conflict (RCON) ($\alpha = 0.742$; CR = 0.749; AVE = 0.504)		
• RCON1: In the project I sometimes have to work around rules and regulations	0.600	Adapted from Babin and Boles (1996)
• RCON2: In the project I received conflicting requests	0.743	
• RCON3: In the project I did things that I knew would be accepted by some but not by others	0.763	
Role ambiguity (RAMB) ($\alpha = 0.816$; CR = 0.826; AVE = 0.557)		
• RAMB1: There were clear and aligned goals for my role in the integrated project management team	0.813	Adapted from Babin and Boles (1996)
• RAMB2: I knew exactly what was expected of me	0.842	
• RAMB3: I knew my duties and responsibilities	0.625	

Note: Confirmatory factor analysis model fit: $\chi^2 = 870.821$, degrees of freedom (DF) = 516, $\chi^2/DF = 1.688$, comparative fit index (CFI) = 0.916, Tucker–Lewis index (TLI) = 0.903, root mean square error of approximation (RMSEA) = 0.066.

(contract type, infrastructure system, stakeholder complexity, and spatial complexity) from Table 1 were tested as control variables in the model to see whether there is a significant impact of these variables on the structural model (see Figure 2).

3.3 | Reliability and validity

The constructs are assessed for reliability and validity in order to validate the measurement model. The reliability is measured using Cronbach's alpha and composite reliability values which assesses the internal consistency. The value must be above the acceptable threshold of 0.7 (Collier, 2020). It can be observed from Table 2 that this criterion is satisfied.

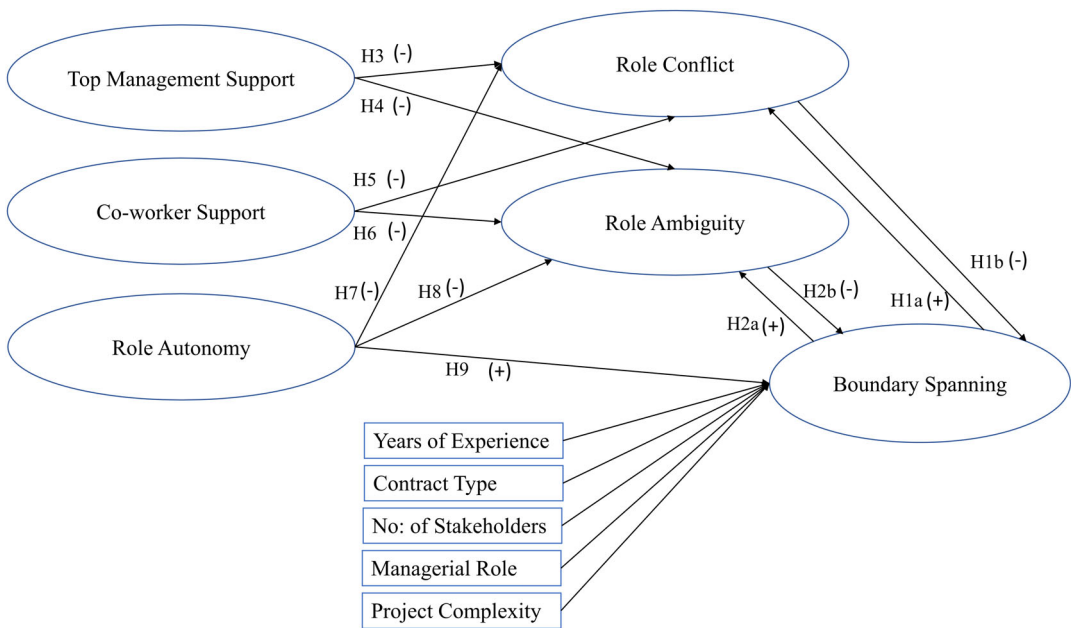


FIGURE 2 The structural equation model with control variables. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/padm.12927)]

The construct validity is assessed through convergent and discriminant validity. Convergent validity is the extent to which an underlying latent construct is reflected by the items. The standardized factor loadings presented in Table 2 are all statistically significant ($p < 0.001$), confirming the convergent validity of the measurement model (Anderson & Gerbing, 1988). To establish discriminant validity, the square root of average variance extracted (AVE) (shown in Table 2) must be higher than the shared variance with other constructs (shown in Table 3); and the lowest acceptable value of AVE is 0.5 (Fornell & Larcker, 1981). These conditions are met, indicating that the constructs are distinct.

4 | ANALYSIS AND RESULTS

The hypotheses were tested using a nonrecursive structural equation model in AMOS, which explores the causal relationships among the latent variables with reciprocal relationships (Wong & Law, 1999). The best model fit was significant ($\chi^2 = 870$, $df = 516$, $p < 0.001$) (Byrne, 2001). The model has a reasonably good overall fit with Tucker–Lewis index (TLI = 0.903), comparative fit index (CFI = 0.916), and root mean square error of approximation (RMSEA = 0.066).

As a first step in nonrecursive modeling, we need to establish that the model is stable by examining if the stability index is between -1 and $+1$ (Arbuckle, 2007). In our case, the stability index is 0.319, indicating that it is within the range. Second, an exogenous variable, referred to as an instrumental variable, is added to deal with endogenous regressor problems in the feedback loop which could create issues for model identification (Wong & Law, 1999). The instrumental variables in this study are top-management support and co-worker support. Third, to satisfy the order and rank conditions, we assign exogenous variables to each variable in the feedback loop.

The descriptive and correlation statistics are presented in Table 3. The means for boundary-spanning activities, role conflict, top-management support, role autonomy, and co-worker support are all above average. This indicates

TABLE 3 Descriptive statistics and correlation between latent variables and control variables.

	Mean	SD	Boundary spanning activities	Role conflict	Role ambiguity	Top-management support	Co-worker support	Role autonomy	Years of experience	Managerial role	Stakeholder complexity	Project complexity	Contract type
Boundary spanning activities	4.95	1.47	1										
Role conflict	4.14	1.33	0.298**	1									
Role ambiguity	2.23	0.86	-0.045	0.194*	1								
Top-management support	5.71	1.01	-0.017	-0.180*	-0.287**	1							
Co-worker support	6.23	0.97	-0.003	-0.093	-0.598**	0.139	1						
Role autonomy	5.63	0.96	0.104	-0.055	-0.429**	0.120	0.657**	1					
Years of experience	2.92	0.997	0.279**	0.211**	-0.067	0.044	0.071	0.159*	1				
Managerial role (project manager = reference)	N/A	N/A	-0.570**	-0.217**	0.052	0.170*	-0.057	-0.036	-0.348**	1			
Stakeholder complexity	2.65	1.749	0.116	0.190*	0.008	-0.024	-0.066	0.076	0.122	0.121	1		
Spatial complexity	1.97	1.213	0.208**	0.206**	0.005	-0.061	0.039	0.183*	0.166*	-0.041	0.262**	1	
Contract type (Design-and-Construct [D&C] = reference)	N/A	N/A	0.120	0.023	-0.007	0.131	-0.002	0.151	0.057	-0.091	0.032	0.022	1

*Correlation is significant at the 0.01 level (2-tailed).

**Correlation is significant at the 0.05 level (2-tailed).

that these phenomena manifest in the respondents' work. In contrast, role ambiguity showed a low mean, indicating that role ambiguity is far less present in comparison to other factors. Among the core variables, boundary-spanning activities is found to have a strong correlation only with role conflict. In the correlation matrix, role ambiguity is correlated negatively with all the organizational conditions. This indicates that role ambiguity is less experienced when there are higher levels of top-management support, co-worker support, and role autonomy. Similarly, role conflict is negatively correlated with top-management support.

It can be observed that more boundary-spanning activities are undertaken when an individual has more years of experience and when the project is characterized by higher spatial complexity. This makes sense as experienced managers can be expected to have gained more experience with spanning boundaries as well; and higher spatial complexity would imply that more organizations are involved as stakeholders in a project (Verweij et al., 2013)—which is also confirmed by the correlation between stakeholder complexity and spatial complexity—making boundary spanning naturally more required in the project. Also, the type of managerial role is seen to influence the degree to which boundary-spanning activities are performed, as shown in Figure 3. We can observe that contract managers, technical managers, and project managers are more involved in boundary-spanning activities than project control managers. Also, the mean boundary spanning by project control managers is below the aggregated mean of all public managers. This could be because project control managers manage the boundaries with the external project environment (e.g., with municipalities or local communities) and do not explicitly focus on the public-private boundary.

From Table 3, we can see that the two role stressors operate quite differently. Role conflict is correlated with both stakeholder and spatial complexity.¹ This is not surprising, as both higher complexities enhance the chances of diverging interests and therefore conflicting role demands for public managers. Similar to boundary-spanning activities, some managerial roles are more strongly correlated with role conflict as well as managerial experience. This is in line with our theoretical model in which we expect that boundary-spanning activities are causing a stronger experience of role conflict.

The results are shown in Table 4. Boundary spanning positively influences role conflict ($\beta = 0.640, p < 0.001$) and role conflict negatively influences boundary-spanning activities ($\beta = -0.440, p < 0.05$). Thus, Hypothesis H1a and H1b are confirmed. This implies that engaging in boundary-spanning activities leads to more role conflict (and the relationship found is strong), and also that as role conflict increases, public managers are less keen to engage in boundary-spanning activities. The relationships between boundary spanning and role ambiguity are insignificant, and

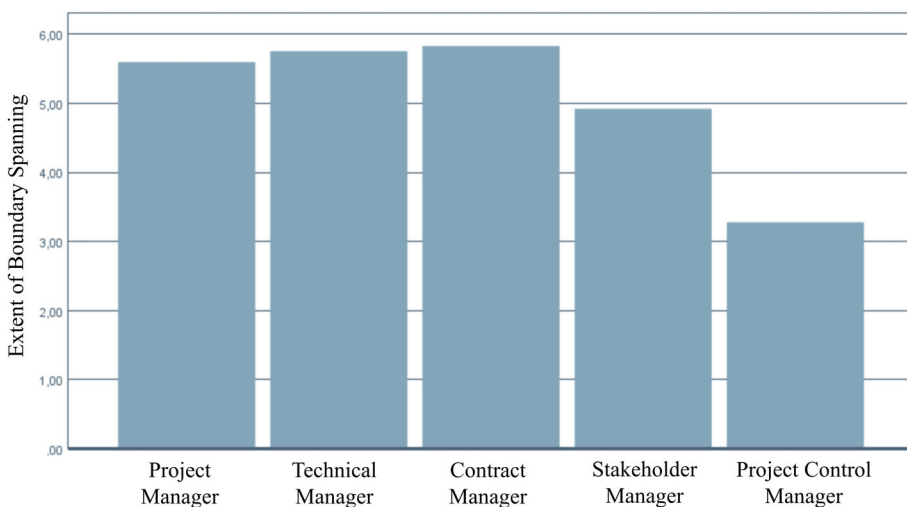


FIGURE 3 Mean score of boundary-spanning activities of the different managerial roles. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pa.12927)]

TABLE 4 Summary of hypothesis testing.

Hypothesis		Standardized coefficients	p-Value
H1a	Boundary spanning → role conflict	0.640	<0.001 (supported)
H1b	Role conflict → boundary spanning	−0.440	0.018 (supported)
H2a	Boundary spanning → role ambiguity	−0.152	0.074 (rejected)
H2b	Role ambiguity → boundary spanning	0.247	0.101 (rejected)
H3	Top-management support → role conflict	−0.205	0.039 (Supported)
H4	Top-management support → role ambiguity	−0.218	0.004 (supported)
H5	Co-worker support → role conflict	−0.026	0.867 (rejected)
H6	Co-worker support → role ambiguity	−0.676	<0.001 (supported)
H7	Role autonomy → role conflict	−0.103	0.521 (rejected)
H8	Role autonomy → role ambiguity	−0.023	0.841 (rejected)
H9	Role autonomy → boundary spanning	0.100	0.431 (rejected)
Control variables			
	Years of experience → boundary spanning	0.077	0.378
	Contract type → boundary spanning	0.030	0.706
	Stakeholder complexity → boundary spanning	0.233	0.007
	Managerial role → boundary spanning	−0.707	<0.001
	Spatial complexity → boundary spanning	0.177	0.040

therefore Hypotheses 2a and 2b are rejected. Hypotheses 3 and 4 are confirmed, indicating that top-management support can help to reduce role conflict ($\beta = -0.205$, $p < 0.05$) and role ambiguity ($\beta = -0.218$, $p < 0.05$). Hypothesis 5 is not supported since there is no significant relationship between co-worker support and role conflict. Hypothesis 6 is supported and indicates that co-worker support significantly decreases role ambiguity ($\beta = -0.676$, $p < 0.001$). Contrary to expectations, role autonomy has no significant relationships with both role stressors (Hypotheses 7 and 8) and boundary-spanning activities (Hypothesis 9).

5 | DISCUSSION AND CONCLUSION

Previous studies have shown that boundary spanners are important in facilitating public–private collaboration processes, and in that way contribute to project performance (Satheesh et al., 2022). They do this by managing information flows, mediating between different interests, building sustainable relationships, and coordinating across public and private organizational boundaries (Delozier & Burbach, 2021; Van Meerkerk & Edelenbos, 2018a; Williams, 2012). Scholars have even suggested that boundary spanners may be important to achieve potential collaborative advantages (Verweij & Satheesh, 2022). However, for boundary spanners to live up to this potential, reducing role stressors—role conflict and role ambiguity—will be pivotal. The present study examined the role stressors experienced by boundary spanning public managers in public–private collaboration on public infrastructure projects, as well as key supporting organizational conditions which could reduce this role stress. The findings show that in particular role conflict negatively affects boundary-spanning activities, and that top-management support is a key supporting organizational condition that can reduce role conflict. In addition, both co-worker support and top-management support can reduce the role ambiguity of boundary spanning public managers.

This study is the first, to our knowledge, to examine the reciprocity between role stressors and boundary-spanning activities. Using a nonrecursive structural equation model, we found that boundary-spanning activities strongly

and positively influence experienced role conflict. This is in line with previous studies conducted in business administration (Lysonski et al., 1988; Stamper & Johlke, 2003). In turn and in addition, a higher level of experienced role conflict reduces boundary-spanning activities. In relation to experiencing a high level of role conflict, we observe a decrease in the number of activities undertaken by the public managers. This is in line with the findings of Wang et al. (2020), who also found a negative influence of role conflict on boundary spanning behavior in the hospitality sector. A possible explanation for this finding is that the difficulty to meet conflicting demands and diverging expectations causes boundary spanners to refrain from performing further boundary-spanning activities, and instead incentivizes them to focus on communication and coordination with their home organization. When experiencing role conflict, they might adopt a behavior of being more reserved and intra-organizational oriented, instead of communicating and relating to the external environment which would be experienced as increasing the complexity and stress of their management task. Given its relatively strong negative influence on boundary-spanning activities, role conflict implies an important risk for managing the collaboration and public infrastructure projects, as it could lead to reducing information exchange, relationship building activities, and alignment between collaborating public and private organizations, while such activities are essential for improving project performance (Brion et al., 2012; cf. Warsen et al., 2019; Satheesh et al., 2022).

Contrary to our expectations, we did not find significant relationships between boundary-spanning activities and experienced role ambiguity. The empirical context of our study might be an important explanation here. The management roles of public project managers are clearly defined and institutionalized within Rijkswaterstaat. Therefore, the managers may know quite well what their goals, duties, and responsibilities are, and what is expected from them—which implies they would experience a lower degree of role ambiguity. This can also be observed from Table 3, where the mean for role ambiguity is clearly much lower than the means for all the other core variables in the study. This situation might be different for public managers in a context where which collaboration and cross-boundary interaction are more emergent and where boundary spanning roles are less clear or emergent (cf. Van Meerkerk & Edelenbos, 2018a; Williams, 2012).

A second key finding of our study concerns the role of several organizational conditions in reducing role stressors. We found especially top-management support to be important for reducing both role conflict and role ambiguity. Top-management support can relieve some pressure from role stressors on boundary spanning public managers. Given the previous finding on the negative impact of role conflict on boundary-spanning activities, this underlines the importance of top-management support in facilitating boundary spanning public managers. Augmenting top-management support by showing commitment, concern, and providing support when needed, helps boundary spanning public managers in dealing with role conflict and contributes to reducing emotional exhaustion (Stamper & Johlke, 2003). This finding underlines the role of top-management, not only as project champions in the start-up of public-private collaboration (Noble & Jones, 2006), but also during the implementation phase, and as facilitators of boundary spanning behavior (Van Meerkerk & Edelenbos, 2018b).

In addition, we find that co-worker support can be particularly relevant for reducing role ambiguity experienced by boundary spanning public managers. Having a supportive team helps in developing a positive orientation towards the organization by increasing trust in co-workers and supervisors (Eslerod et al., 2017). The realization that help is available when needed increases the self-efficacy of a boundary spanner, thereby lessening role ambiguity (Eslerod et al., 2017). This can help boundary spanners in making expectations clearer (towards them) and in this way can also lead to more perceived self-efficacy about their spanning activities. However, we did not find a significant effect of co-worker support on role conflict. This implies that co-worker support, according to our study, does not change or help in dealing with experienced stress due to conflicting demands from different intra- and extra-organizational stakeholders. To alleviate role conflict, especially top-management support is needed.

Contrary to our expectations, we did not find significant relationships between role autonomy on one hand and experienced role stress and boundary-spanning activities on the other hand. Again, this might be explained by the specific context of the study. When working in a project-oriented organization, the project team is seen as a reference point. In developing public infrastructure projects, the expertise of many individuals is combined and this

creates a dependency to achieve the anticipated outcome. This presents an opportunity for further research to study the project team dynamics that affect individual boundary spanning behavior. Probably, the management of public infrastructure projects in the Netherlands is much more a team effort where roles are clearly defined and delineated, leaving less room for taking project-tailored or individualized management approaches. In this context, this could make role autonomy a mechanism less available to make a difference in terms of dealing with role stressors. Further in-depth research should investigate how plausible this explanation really is.

This study contributes to systematically testing the role of supporting factors—organizational factors—and hindering factors—role stressors—in relation to boundary-spanning activities in public–private collaborations. A future study could assess boundary spanning roles and how these are affected by supporting organizational conditions and role stressors, the idea being that activities come together in roles and that activities may be performed differently depending on roles (Van den Brink et al., 2019). Future research could also explore the various coping mechanisms and the role of personality traits and intrinsic motivation of boundary spanning public managers in dealing with role stressors.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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ENDNOTE

- ¹ The control variables having significant correlations with role conflict were tested in the structural model. However, they did not have significant parameter estimates and as a result, these relationships are not included in the final model.

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